

# How to use the Super Steward Kit to conduct an investigation

## 1 Collect a sample from nature in your local environment

- This can be anything in nature, for example: Soil, water, flower, leaves, traces of animals, or anything else.
- Put a small amount of the sample in a dish or tube (you choose).

## 2 Study the samples under the Dissecting Microscope

- Follow the instructions on How to use a Dissecting Microscope.



## 3 Give each sample a score

Use the scale bar below to give each sample a score from 1 to 10 (Natural, In-between or Un-natural). Refer to the examples to assist you in completing this task.



### Un-Natural sample



**Things made by humans or machines**

Examples: Toys, rubbish, roads, electrical items

### In-Between Sample



**Things containing both natural and unnatural things or things that started as natural but were changed by humans/machines.**

Examples: Water or soil with small bits of plastic or rubbish

### Natural sample



**Things that come directly from the environment**

Examples: Clean water, feathers, leaves and flowers

## 4 Record your Results

- Record the details of the samples you collected and their score in the Data Collection Sheet.
- Give reasons for each score in the last column.

# How to analyse the data and reflect on the results

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## Use your results to calculate the Average Score:



- Add the totals on your data table (in the blue and green boxes)

| Sample Type           | Location                                 | What do you see/observe?                                 | Score<br>Give it a number using the Score bar above | Reason for score                                                      |
|-----------------------|------------------------------------------|----------------------------------------------------------|-----------------------------------------------------|-----------------------------------------------------------------------|
| Dish of water         | Small water puddle near the science room | Insects, clear liquid, sticks and possible microplastics | 8                                                   | The sample is mostly natural but could possibly have harmful plastics |
| Piece of a K-time bar | Playground                               | Unfinished snack that someone didn't put in the bin      | 3                                                   | Contain fruit but made with sugars and other non natural ingredients  |
| Paper (rubbish)       | Near the lockers                         | Scrunched late pass                                      | 5                                                   | Comes from trees but made and used by humans                          |
|                       |                                          | <b>Total Number of Samples:</b><br>3                     | <b>Total Score:</b><br>8+3+5 =16                    |                                                                       |

- Calculate the Overall rating using the equation (copy from the blue and green boxes):

$$\text{Average Score} = \frac{\text{Total Score}}{\text{Total Number of Samples}}$$

$$\text{Average Score} = \frac{16}{3}$$

$$= 5.33$$

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## What does your result tell you? Think back to your original question (from the 'Ask' Stage)

The average score of the items in this environment is 5.33 out of 10. Overall the score for this environment is mostly "in-between", so not entirely natural or unnatural.